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Comparing Team Selection and Seeding for
the 2011 NCAA Men's Basketball
Tournament

Kathy L. Gray, *California State University, Chico*
Neil C. Schwertman, *California State University, Chico*

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Abstract

The men's NCAA basketball tournament is a popular sporting event often referred to as "March Madness." Each year the NCAA committee not only selects but also seeds the tournament teams. Invariably there is much discussion about which teams were included and excluded as well as discussion about the seeding of the teams. In this paper, we propose an innovative heuristic measure of team success, and we investigate how well the NCAA committee seeding compares to the computer-based placements by Sagarin and the rating percentage index (RPI). For the 2011 tournament, the NCAA committee selection process performed better than those based solely on the computer methods in determining tournament success.

KEYWORDS: NCAA, men's basketball, seeding, RPI, Sagarin

INTRODUCTION

Each March the NCAA sponsors basketball tournaments featuring teams from various universities to play for the National Championship. While the women's Division I tournament is growing in popularity, it still is overshadowed by the men's event. The tournaments often sprout office pools and are frequently referred to as "March Madness". Furthermore, the men's tournament has been the subject of several statistical analyses, see for example, Carlin (1996), Caudill and Godwin (2002), Schwertman et al. (1991), Schwertman et al. (1996), Smith and Schwertman (1999) or West (2008). The focus of these papers is to predict team's success in the tournament based on their seed position or other measures of team strength. In this paper, however, instead of concentrating on predicting success, the interest is how well a team's tournament success correlates with their seeding by the NCAA committee and two seed placements schemes based on computer ratings.

In designing the tournament, the NCAA uses a panel of experts to select and seed the teams to compete for the National Championship. Division I conference champions or their conference tournament winners are automatically included in the tournament to give a broad representation of the basketball talent throughout the United States. Besides the automatic qualifiers, about half of the tournament teams are "at-large" selections based on their record. Of these, there are always several teams which are "on the bubble", that is, inclusion in the tournament is questionable and could go either way. To include more "bubble" teams, in 2011 the NCAA expanded the field to 68 teams. Sixty teams were assigned a seed position while the other eight teams played a preliminary game, four played for a 16th seed, two each played for an 11th and 12th seed. These preliminary games are considered as an additional regular season game where the eight teams are competing for inclusion in the tournament. Thus the final 64 teams are now on an equal basis and only the play of the final 64 teams after the preliminary games are used in the analysis. Following the selection committee announcement of the teams and seeding there is a great deal of controversy about which teams were included and excluded and also the seed placements.

The objectives of this paper are first to propose an innovative simple heuristic measure of a team's success in the tournament, second, to examine how well the seed placement correlates with the new measure of team's success and finally to determine if seeding based only on the computer ratings correlate better with team success.

The computer ratings used are the Sagarin and RPI scores. The Sagarin computer ratings which appear regularly in *USA Today* is by Jeff Sagarin and incorporates two rating systems. The first, the Elo Chess used for rating chess players is adapted to basketball and uses only wins and losses while the second,

called the predictor system, uses the scores of each game. The exact methodology is a well kept secret. The second computer ratings is the RPI which stands for Rating Percentage Index and is a weighted average with 25% weight given to the team's winning percentage, 50% to their opponent's winning percentage and 25% to their opponent's opponent's winning percentage.

SEEDING OF THE 64 TOURNAMENT TEAMS

Figure 1 is the general pattern used by the NCAA to pair the seeds. The placement of teams are based on the NCAA Selection Committee perceived strength or, in the case of computer ratings, on the computer calculated strengths with the first four being the number 1 seeds and the next four being the number 2 seeds, etc. The two seed placements of the 64 teams in the 2011 tournament, based on the Sagarin and RPI scores, as well as the actual NCAA seeding are provided in Table 1.

Figure 1: NCAA regional basketball tournament pairings

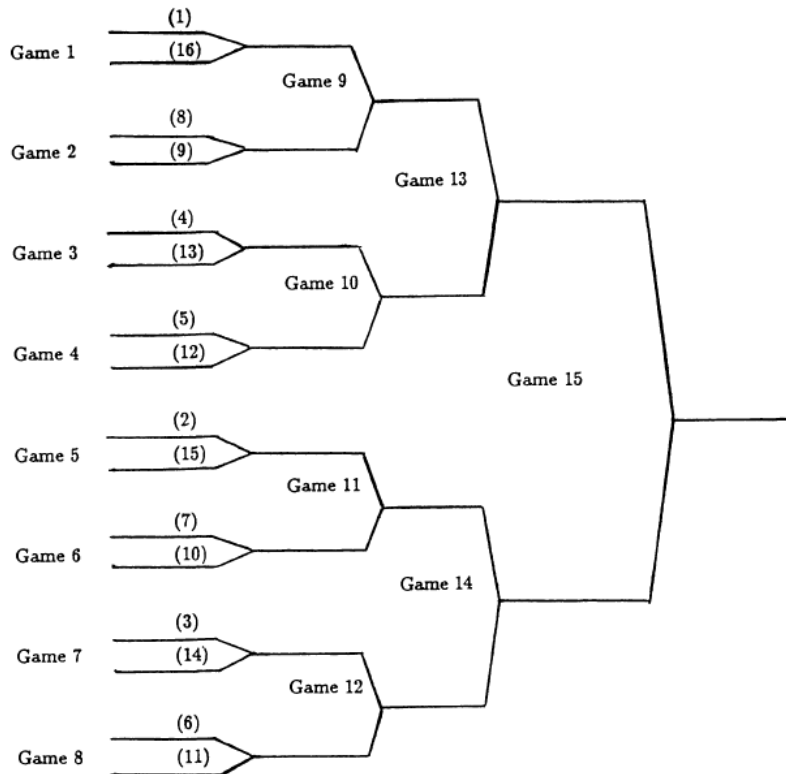


Table 1: Seeding, Sagarin and RPI scores and sum of success points of the final 64 teams in the 2011 Men's Basketball Tournament

School	NCAA seeding	Sagarin Seeding	RPI Seeding	Sagarin raw score	RPI raw score	NCAA Success Points Sum	Sagarin Success Points Sum	RPI Success Points Sum
Ohio St.	1	1	1	95.73	0.668	5.1652	5.3075	5.6193
U.T. S.A.	16	16	16	65.52	0.4837	1.4996	1.4996	1.4996
G. Mason	8	8	7	84.89	0.6052	3.0186	3.2658	3.0924
Villanova	9	6	8	85.72	0.5905	1.2624	1.2624	1.3344
W. Va.	5	5	5	86.9	0.6177	2.7546	2.9458	3.0661
Clemson	12	11	12	83.52	0.5724	1.4576	1.4576	1.4576
Ky	4	3	2	89.11	0.6429	9.5635	9.0325	9.5909
Princeton	13	13	10	79.9	0.5821	1.6694	1.8114	2.0085
Xaxier	6	9	6	84.29	0.6081	0.9667	1.1258	0.8775
Marquette	11	8	13	84.98	0.5692	5.4286	4.8784	5.1475
Syracuse	3	3	5	89.91	0.6181	2.3782	2.5576	2.2776
Indiana St.	14	14	14	76.09	0.5509	1.4610	1.4610	1.2553
Washington	7	7	9	85.24	0.5892	3.2817	2.8237	3.2200
Georgia	10	12	11	81.93	0.5774	1.365	1.365	1.2254
UNC	2	4	2	88.76	0.6435	5.6619	6.0960	5.8043
Long Island	15	14	13	77.11	0.5541	1.7733	1.4739	1.7733
Duke	1	1	1	92.96	0.6588	4.4861	4.2692	4.2086
Hampton	16	16	16	71.7	0.4978	1.2647	1.2647	1.2647
Michigan	8	10	11	83.91	0.5764	4.5147	4.2197	4.5147
Tennessee	9	12	9	82.06	0.5889	0.8106	0.7314	0.696
Arizona	5	6	6	86.26	0.613	7.6579	7.8181	7.7680
Memphis	12	13	8	80.35	0.5948	1.5692	1.4737	1.4737

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Texas	4	2	4	90.52	0.6322	2.6930	2.6510	2.6510
Oakland U.	13	12	12	81.53	0.5756	1.6467	1.9812	1.6467
Cincinnati	6	5	8	87.56	0.5907	3.0123	3.2205	3.0398
Missouri	11	7	9	85.06	0.5882	1.2220	1.3012	1.0912
U. Conn.	3	4	4	88.9	0.6297	10.828	11.225	11.207
Bucknell	14	13	14	78.01	0.5506	1.2037	1.1093	1.1093
Temple	7	8	8	84.78	0.5967	3.1311	3.4002	3.4610
Penn St.	10	11	10	83.17	0.5863	1.3847	1.3101	1.3101
S. Diego St.	2	1	1	91.9	0.6591	4.5833	4.3648	4.3648
N. Colorado	15	15	15	75.22	0.536	1.5287	1.7978	1.7978
Kansas	1	1	1	94.54	0.6757	5.3145	5.6423	5.3314
Boston U.	16	16	15	72.17	0.5173	1.7998	1.7998	1.7998
UNLV	8	5	7	86.86	0.6064	1.0888	1.2128	0.9852
Illinois	9	7	11	85.22	0.577	3.5668	3.8729	3.6575
Vanderbilt	5	8	7	84.94	0.6042	1.0570	1.2263	1.166
Richmond	12	9	10	84.13	0.5815	4.9201	4.5819	4.6624
Louisville	4	4	4	89.03	0.6205	1.0360	0.9878	0.9878
Morehead	13	14	14	76.3	0.5526	2.5723	2.7030	2.6565
Georgetown	6	4	3	87.56	0.6342	0.8778	0.7969	0.8362
Va. C.U.	11	13	12	79.52	0.5738	9.5970	9.6521	9.8986
Purdue	3	3	3	90.25	0.6351	2.4555	2.2980	2.4111
St. Peter's	14	15	14	75.09	0.5404	1.2404	1.2404	1.2404
Texas A&M	7	9	7	84.7	0.5984	1.0693	1.0175	0.9693
Florida St.	10	11	12	83.22	0.5739	5.3614	5.0895	5.3072
Notre Dame	2	2	2	90.49	0.6394	2.1578	2.1104	2.0663
Akron	15	15	15	75.11	0.5354	1.6873	1.6873	1.6873

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Pittsburgh	1	2	3	91.77	0.6375	2.6553	2.4670	2.5872
U.N.C. Ash	16	16	16	73.06	0.5071	1.6851	1.4328	1.2922
Butler	8	11	9	82.53	0.5885	10.685	9.6925	9.9575
O.Dominion	9	10	6	83.71	0.6115	1.3060	1.1213	1.2393
Kansas St.	5	7	5	85.35	0.6137	2.7908	3.3653	3.3340
Utah St.	12	6	5	86.52	0.6182	1.5007	1.3302	1.5007
Wisconsin	4	3	4	89.31	0.6229	4.2381	4.1562	4.1770
Belmont	13	9	13	84.4	0.5684	1.392	1.5104	1.392
St. John's	6	6	6	85.87	0.6116	0.9577	1.0064	0.8693
Gonzaga	11	10	13	83.69	0.5721	3.2442	3.3977	3.3977
BYU	3	2	2	90.84	0.6455	4.5202	4.1813	4.1434
Wofford	14	14	15	77.11	0.5346	1.6723	1.8542	1.8542
UCLA	7	12	10	82.44	0.5815	3.1022	2.6855	2.8600
Mich. St.	10	10	11	83.99	0.5802	1.3914	1.0767	1.1877
Florida	2	5	3	87.3	0.6391	6.4854	6.1682	6.3436
UC S. Barb	15	15	16	73.53	0.4996	1.3421	1.0400	1.2104

MEASURING TOURNAMENT SUCCESS

The easiest method of determining tournament success is to count the number of wins. Thus, a top seed win over a bottom seed would count the same as the other way around. In addition, the relative strength of the team and the closeness of the game are not considered and hence there is no measure of the relative successes within the groups with no wins, one win, etc.. A second simple method is to count a first round win 1 point, a second round 2 points, etc.. This assumes that as the tournament progresses the competition is tougher. Similar to just counting wins, this method does not consider the closeness of the score nor provides the relative success within the teams grouped by number of wins and, at best, only roughly incorporates the quality of the competition.

As an alternative, we suggest a heuristic measure of success. Specifically, the number of points a team scores in a game is an indication of how well they did

on offense and the inverse of what their opponent scored is a measure of how well they did on defense. Thus, the ratio of their score to the opponent's score incorporates the teams overall performance in that game. This still does not measure the quality of the competition. To make this adjustment we suggest using the z-scoring introduced by Schwertman et. al (1991) to measure the relative strengths of the teams. Using their procedure, the average percentiles for seeds 1 through 16 are z-scored. For example, the top seeds are ranked 1, 2, 3 and 4 and the median of these numbers is 2.5. Since there are 345 teams in Division I and using the continuity correction, the percentile is $100(1-2.5/345.5) = 99.276\%$ tile which equates to $Z=2.445$. For the number 2 seeds the rankings are 5, 6, 7 and 8, the median is 6.5, the percentile is $100(1-6.5/345.5) = 98.119\%$ tile and $Z=2.079$. Table 2 gives the z-score for each seed.

Table 2: Z-scores for the 16 seeds

Seeding	1	2	3	4	5	6	7	8
Z-score	2.445	2.079	1.875	1.728	1.611	1.513	1.428	1.351
Seeding	9	10	11	12	13	14	15	16
Z-score	1.282	1.219	1.160	1.105	1.053	1.004	.957	.912

This measure may still give the losing team more success points than the winning team but the winning team's advantage is that they continue playing and are afforded the opportunity to acquire more points. For example, in the first round Kentucky, a 4 seed, defeated Princeton, a 13 seed, 59 -57 and their success points are $(59/57) 1.053 = 1.08995$ while Princeton's success points are $(57/59) 1.728 = 1.66942$. As illustrated, the measure of success places the stronger seeds at a distinct disadvantage in the early rounds since they face weaker opposition and the z-scores of the competition are small but, as in the case of Kentucky, the opportunity for further play allows for future success. Kentucky was the East Region champions and was fourth in total points. Table 1 includes the total success points for the final 64 teams.

INVESTIGATING THE QUALITY OF SEED PLACEMENT AND THE RELATIONSHIPS

The summary of the success points of the final 64 tournament teams, their seeding by NCAA, Sagarin and RPI scores and their raw Sagarin and RPI scores are provided in Table 1. The correlation provides a measure of the strength of the linear relationship between the various variables. We investigated the relationships between the sum of a team's success points calculated and the three seed placements for NCAA, Sagarin and RPI. Table 3 displays the correlations

between seeding and success point sums. The strongest associations were between the NCAA success point sum and the three seeding types, -.397, -.411, -.399, respectively. These correlations were slightly higher than the correlation between Sagarin success point sum and the different seeding types and also slightly higher than the correlation between RPI success point types and the three seeding types. Based on the magnitude of the correlation coefficients in Table 3, it appears that the seedings given by the NCAA, Sagarin, and RPI were not a good predictor of a team's success points.

Table 3: Correlation between seeding and sum of success points

	NCAA Point Sum	Sagarin Point Sum	RPI Point Sum
NCAA Seeding	-.397	-.319	-.315
Sagarin Seeding	-.411	-.332	-.321
RPI Seeding	-.399	-.320	-.310

Table 4 displays the correlations between the success points calculated for NCAA, Sagarin, and RPI. As would be expected, a very strong association between the three success point measures based on the three different seed rankings indicating little difference between the three types of seed placement.

Table 4: Correlation between success point sums for NCAA, Sagarin, and RPI

	NCAA Point Sum	Sagarin Point Sum	RPI Point Sum
NCAA Point Sum	1.0	.994	.996
Sagarin Point Sum	.994	1.0	.998
RPI Point Sum	.996	.998	1.0

Table 5: Correlations between seedings

	NCAA Seeding	Sagarin Seeding	RPI Seeding
NCAA Seeding	1.0	.915	.934
Sagarin Seeding	.915	1.0	.922
RPI Seeding	.934	.922	1.0

Table 5 gives the correlations between the NCAA, Sagarin and RPI seed rankings of the teams. Similarly, there is a strong association between the seeding rankings themselves indicating that there is little difference in any particular seed placement method. The association between each team's success points based on the NCAA seeding was increased by using the raw scores from the Sagarin and RPI rather than the seeding based on these ratings. The Sagarin absolute correlation increased from .319 to .3303 and the RPI absolute correlation increased from .315 to .3531 suggests that the additional information in the raw scores only marginally enhances the association with the success scores.

Table 6: Total number of upsets by region

Round	Region	NCAA	Sagarin	RPI
1	East	1	1	1
	West	0	0	0
	Southwest	5	5	4
	Southeast	1	4	2
2	East	1	1	1
	West	1	1	1
	Southwest	2	2	2
	Southeast	1	2	2
3	East	1	1	1
	West	2	2	2
	Southwest	1	1	0
	Southeast	1	2	2
4	East	1	0	0
	West	0	0	0
	Southwest	1	1	1
	Southeast	1	1	1

Another measure of the accuracy of the various placements of seeds is the upsets that occur in the tournament. Table 6 lists the number of upsets by region in each round. Clearly in 2011 the Southwest region was upset prone with 9 out of the 15 games considered upsets while the other regions had either 3 or 4 upsets. The total number of upset were: 20, 24 and 21, respectively, for the NCAA, Sagarin and RPI seed placements. Weighting the upsets by the difference in the seeding, the total weighted upset scores were 98, 112 and 117 for NCAA, Sagarin and RPI, respectively.

TEAMS OMITTED AND THEIR ASSIGNED SEED

The NCAA is obligated to include the conference winners or their tournament champions in the field of tournament teams which necessitates omitting some deserving teams. Table 7 lists the teams in the 2011 tournament that were not included but who's Sagarin or RPI ranking indicate that their record was sufficient to be included in the tournament.

Table 7: List of teams with a high Sagarin or RPI ranking but not selected for tournament

Team	Sagarin Seeding	RPI Seeding
Virginia Tech	11	Not ranked
St. Mary's	11	13
New Mexico	12	Not ranked
Colorado	13	16
Maryland	13	Not ranked
U. Alabama Birmingham*	14	8
Northwestern	14	Not ranked
Washington State	14	Not ranked
Minnesota	15	Not ranked
Boston College	15	15
Nebraska	15	Not ranked
Alabama	15	Not ranked
Cleveland State	16	11

Harvard	16	10
Wichita	16	16
Missouri State	Not ranked	10
Colorado State	Not ranked	12
Texas El Paso	Not ranked	14
Oklahoma State	Not ranked	15
Marshall	Not ranked	15
Southern Mississippi	Not ranked	16

*University of Alabama Birmingham lost to Clemson for an 11th seed in the preliminary games.

CONCLUDING COMMENTS

While it is unlikely that the Selection Committee will ever achieve perfection, during the first year of the 64 team format (1985) there were no upsets (lower seeds beating upper seeds) in the Eastern region which is considered perfect placement for this region. This analysis of 2011 tournament data shows that the Selection Committee in 2011 was quite effective in the seeding of the tournament and that basing the seeding entirely on computer ratings was not an advantage since, of the three placements, the NCAA seeding had the strongest correlation with team success points and the fewest upsets. The incorporation of some subjectivity into their seeding appears to be advantageous. The committee is able to make an adjustment, for example, if a key player is injured or unable to play for some reason. From this analysis of the data for the 2011 tournament, there is ample evidence that NCAA selection committee was proficient in the selection and seeding of the tournament teams.

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